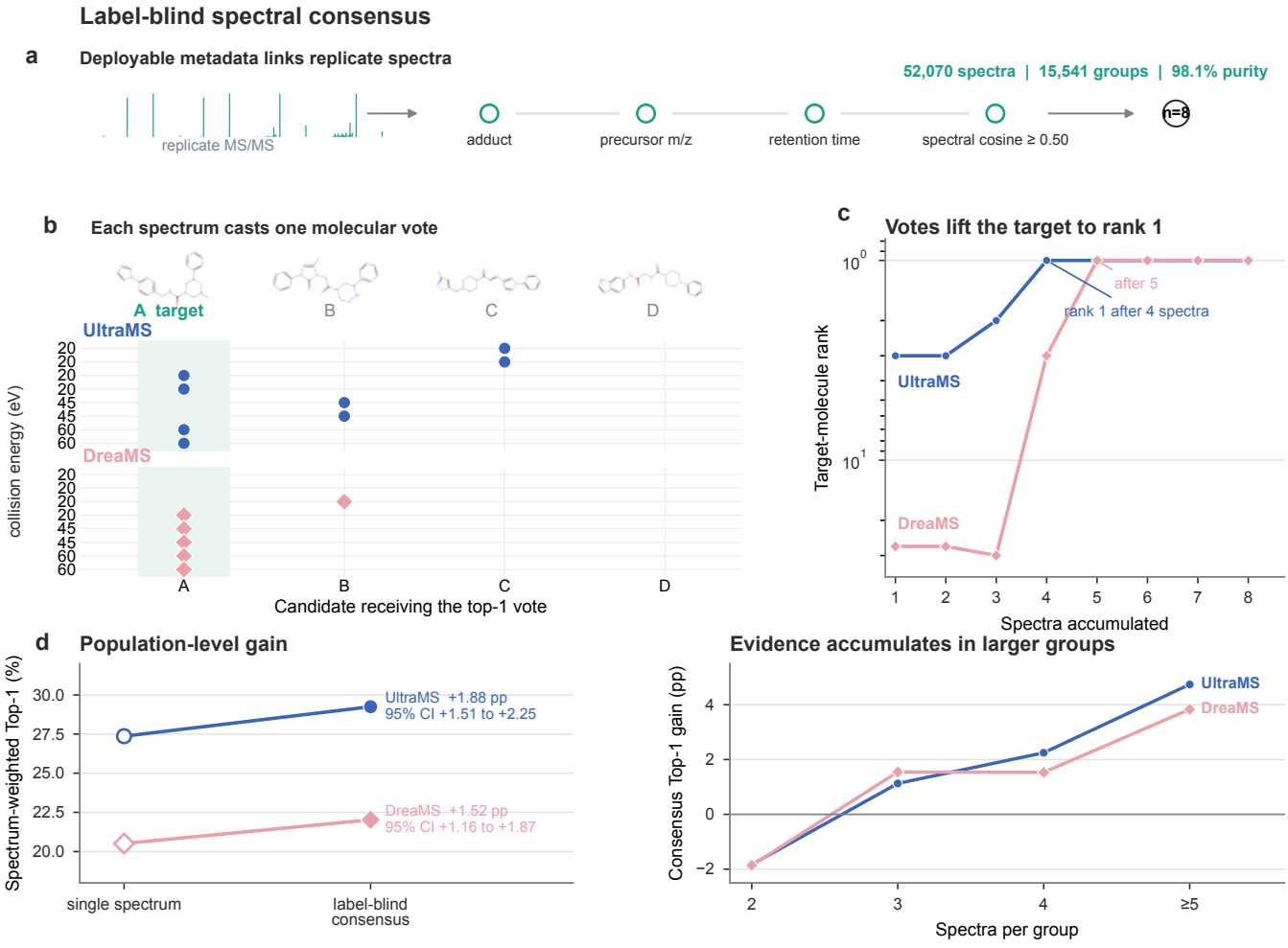

Label-blind spectral consensus
a
Deployable metadata links replicate spectra
52,070 spectra | 15,541 groups | 98.1% purity
replicate MS/MS
adduct
precursor m/z
retention time
spectral cosine ≥ 0.50
n=8
b
Each spectrum casts one molecular vote
A target
B
C
D
UltraMS
DreaMS
collision energy (eV)
Candidate receiving the top-1 vote
c
Votes lift the target to rank 1
Target-molecule rank
after 5
rank 1 after 4 spectra
UltraMS
DreaMS
Spectra accumulated
d
Population-level gain
Spectrum-weighted Top-1 (%)
UltraMS +1.88 pp
95% CI +1.51 to +2.25
DreaMS +1.52 pp
95% CI +1.16 to +1.87
single spectrum
label-blind consensus
Evidence accumulates in larger groups
Consensus Top-1 gain (pp)
UltraMS
DreaMS
Spectra per group